

Microsatellite DNA analysis of genetic effects of harvesting in old-growth eastern white pine (*Pinus strobus*) in Ontario, Canada

O. P. RAJORA,* M. H. RAHMAN,* G. P. BUCHERT†‡ and B. P. DANCİK*

*Department of Renewable Resources, University of Alberta, Edmonton, Alta, Canada T6G 2H1, †Ontario Forest Research Institute, Ontario Ministry of Natural Resources, 1235 Queen Street East, PO Box 969, Sault Ste. Marie, Ont., Canada P6A 5N5

Abstract

Microsatellite DNA markers from 13 simple sequence repeat (SSR) loci were used to compare genetic diversity between preharvest pristine and postharvest residual gene pools of two adjacent virgin, old-growth (≈ 250 years) stands of eastern white pine (*Pinus strobus* L.) in Ontario. There was concurrence in genetic diversity changes in the postharvest gene pools of the two stands. The total and mean numbers of alleles detected in each stand were reduced by $\approx 26\%$ after tree density reductions of $\approx 75\%$. Approximately 18 and 21% of the low-frequency ($0.25 > P \geq 0.01$) alleles and 76 and 92% of the rare ($P < 0.01$) alleles were lost from residual stands A and B, respectively, after harvesting. Multilocus gametic diversity was reduced by 38 and 85% and genotype additivity by $\approx 50\%$ in the residual stands after harvesting. Latent genetic potential of each stand was reduced by $\approx 40\%$. Although heterozygosity was reduced (1–5%) in the postharvest residual stands, the reductions were not substantial and not comparable to those using other genetic diversity measures. The reductions in genetic diversity measures were slightly higher than those theoretically expected in postbottleneck populations according to drift theory. In the absence of substantial gene migration that could ameliorate the genetic losses, the ability of the postharvest white pine gene pools to adapt to changing environmental and disease conditions may have been compromised. The microsatellite DNA results for genetic effects of harvesting in old-growth eastern white pine stands were similar to those that we reported earlier from allozyme analysis (Buchert *et al.* 1997). The results indicate that silvicultural practices should ensure that the gene pools of remaining pristine old-growth stands are reconstituted in the regenerating stands.

Keywords: gene conservation, genetic diversity, *Pinus strobus*, silvicultural practices, simple sequence repeats, sustainable forest management

Received 21 June 1999; revision received 26 October 1999; accepted 26 October 1999

Introduction

Genetic diversity provides raw material for adaptability and evolution of species and individuals, especially in forest tree populations that are undergoing fundamental population changes as a result of natural or anthropogenic perturbances (Gregorius *et al.* 1985; Müller-Starck 1985;

Ledig 1988; Namkoong 1991; Müller-Starck *et al.* 1992). Tree populations that have sustained genetic losses will probably be more susceptible to environmental or disease-related reductions in productivity, fitness and health (Müller-Starck 1985; Bergmann & Scholz 1987; Bergmann *et al.* 1990; Oleksyn *et al.* 1994; Raddi *et al.* 1994). Therefore, genetic diversity is a key component for the sustainability of forest resources. Forest management practices based on partial or clear-cut harvesting followed by natural and/or artificial regeneration can significantly impact genetic diversity, population genetic structure and genetic processes in subsequent forest populations.

Correspondence: Om P. Rajora. Tel: +1-780-492-4020; Fax: +1-780-492 4323; E-mail: Om.Rajora@ualberta.ca

‡Present address: The Algoma Institute, 330 North 100 West, Provo, UT 84601, USA.

However, the impact of various harvesting practices on genetic diversity and genetic processes are very poorly understood. Information on the genetic impact of alternative silvicultural practices is quite rudimentary (e.g. Knowles 1985; Neale 1985; Rajora 1999).

Eastern white pine (*Pinus strobus* L.), is one of the most important softwood timber species in eastern Canada and the northeastern United States. This species has been an integral part of the history and economic development of eastern North America. Once a predominant tree species in forest landscapes of the Great Lakes region of North America, eastern white pine has undergone fundamental changes in population structure and dynamics over 150 years of heavy exploitation (Buchert 1994). Extensive stand harvesting and high grading (selective harvesting of only the best trees from the stand) of eastern white pine has resulted in forest fragmentation and reduction of its population size (Buchert 1994). This population reduction may have resulted in genetic changes to local gene pools (Barnes 1989; Ledig 1992). Ontario and Quebec still have some areas of pristine old-growth white pine forest (Perera & Baldwin 1993), which constitute most of the remaining undisturbed gene pools of the species. As white pine is the keystone species (G. P. Buchert, O. P. Rajora, personal observations) in old-growth white pine ecosystems, its continued presence is essential for many floral and faunal associations in these ecosystems (Barnes 1989; Carleton & Gordon 1992; Welch *et al.* 1992).

Intact, pristine old-growth white pine stands, as in Ontario and Quebec, provide a unique opportunity to develop benchmarks for the levels of genetic diversity inherent in these virgin white pine stands and genetic effects of harvesting, necessary for the development of biologically sustainable silvicultural practices (Ledig 1992). In a finite population, genetic diversity is expected to be reduced as population size decreases (Frankel & Soule 1981). Eastern white pine is primarily regenerated under the shelterwood and seed-tree-cut harvesting systems. The shelterwood harvesting system requires a series of partial cuttings over a period of years in mature stands. The residual trees produce seeds for natural regeneration and provide shelter for the young seedlings. The seed-tree-cut system involves logging of the mature tree stands in one cut, leaving a small number of seed trees singly or in a small group as a seed source for regenerating the stand. An understanding of the impact of various harvesting systems on local gene pools will facilitate the monitoring of such practices to ensure their effectiveness in maintaining long-term ecosystem sustainability and genetic integrity (Mosseler & Rajora 1998). Genetically sound silvicultural practices need to be developed for sustainable management and conservation of forest genetic resources.

In an earlier study we examined the effects of partial harvesting on genetic diversity at 54 allozyme gene loci in a total census of two adjacent old-growth white pine stands, by comparing intact preharvest and residual post-harvest gene pools (Buchert *et al.* 1997). Genetic diversity was found to be reduced by 25–50% in the postharvest residual gene pool, which serves as a seed source for natural regeneration of the forest. As allozymes are generally most clearly resolved and accurately interpreted by using the haploid megagametophyte tissue of seeds in conifers, and there is a limitation on the number of allozyme loci that could be clearly resolved in foliage, seeds are generally used for allozyme analysis in pines. However, trees that are fecund could only be analysed in this manner. Also, to determine the genetic effects of harvesting, postharvest young regenerants need to be analysed. If allozymes show developmental and tissue-specific differences, direct comparisons cannot be made using different tissue types. Although our previous study was based on a total reproductive census of the stands, trees that did not produce seed in the year of sampling could not be genotyped allozymatically because we used seed megagametophyte tissue for analysis. As allozymes have certain limitations and the simple sequence repeat (SSR)/microsatellite DNA markers offer several advantages, we have used microsatellite DNA markers in the present study to examine the genetic effects of harvesting and to verify our allozyme results by comparing the same pre-harvest and residual postharvest gene pools in two adjacent old-growth eastern white pine stands in Ontario.

Materials and methods

White pine stands and trees, silvicultural treatments and sampling

The study was conducted in two virgin old-growth white pine stands (A and B), located ≈ 100 km north of Sault Ste. Marie, Ontario, as a part of the Galloway Lake Old Pine Area ($\approx 47^{\circ}13'$ N latitude, $83^{\circ}56'$ W longitude). The selected stands consisted of a canopy of white pine with an average age of 250 years, with mixed conifer and hardwood understory. The study stands were part of a 250-ha area designated for timber harvesting in 1992. A partial cutting system was prescribed with the objectives of extracting timber, encouraging natural regeneration and maintaining old-growth structural characteristics in residual stands. This was achieved by the silvicultural marking prescription calling for residual spacing to be at 15 m between residual trees in stand A and at 20 m between trees in stand B. The details of the composition and physical parameters of the study stands and silvicultural treatments have been described previously (Buchert *et al.* 1997).

Table 1 Simple sequence repeat (SSR) loci, repeat sequence, number of alleles per locus and observed size of alleles in eastern white pine

Locus	Repeat	No. of alleles	Allele sizes (bp)
RPS1b	(AC) ₁₀	6	217–191
RPS2	(AC) ₁₅	10	171–145
RPS12	(AC) ₁₇	21	195–153
RPS20	(AC) ₁₆ (AT) ₆	16	162–100
RPS25b	(AC) ₁₇ AG(AT) ₉	10	125–101
RPS34b	(AC) ₁₄	6	151–141
RPS39	(AC) ₁₇	6	180–160
RPS50	(AC) ₁₇	13	184–152
RPS60	(AC) ₁₉ (AT) ₇	17	279–247
RPS61	(AC) _n	1	179
RPS118b	(AC) ₂₃	19	166–130
RPS119	(AC) ₁₀ (AT) ₅	2	205–203
RPS127	(AC) ₁₀ (AT) ₅	3	195–191

After each stand was marked for harvest, all white pine trees (132 in stand A and 106 in stand B) were numbered and mapped. Foliage samples were collected from all the white pine trees in each stand. Therefore, our study was based on a total census of the individuals within each stand. After harvest in October 1992, 30 of the original 132 trees were retained in stand A, whereas 23 of the 106 trees were retained in stand B. These residual trees were identified and used for the postharvest gene pool analysis. The postharvest natural regeneration was monitored. No adequate eastern white pine natural regeneration was present until 1996 that could be sampled for genetic diversity assessment.

Microsatellite DNA analysis of genetic diversity

DNA was extracted from the foliage of 120 individual trees from Stand A and from the foliage of 102 individual trees from stand B. Foliage samples of the remaining 12 trees from stand A and four trees from stand B were lost. Genotype and genetic variability of the 222 trees were determined by using 13 eastern white pine SSR primer pairs (Map Pairs™; Research Genetics) found to be suitable. A total of 23 primers flanking microsatellite loci of dinucleotide repeats (Echt *et al.* 1996) were screened repeatedly. The methodology for both the generation of microsatellite markers and the visualization of these markers was developed and optimized. The microsatellite markers were visualized using vertical denaturing gel electrophoresis and silver staining. With refinement and optimization of the protocol, 13 of a total of 23 primer pairs were found to produce reliable single and scoreable PCR products, which were used for examining microsatellite DNA variation in the white pine trees. Allelic

segregation for all the polymorphic microsatellite DNA loci used in this study (Table 1) was tested previously in six haploid megagametophytes, each of 16 white pine trees (Echt *et al.* 1996), and three of these loci (*RPS12*, *RPS50* and *RPS118b*) have been genetically mapped in white pine (Echt & Nelson 1997).

Polymerase chain reaction amplification of microsatellites

Polymerase chain reaction (PCR) reaction mixtures contained 10 ng of DNA template in 10 µL of reaction buffer: 10 mM Tris-HCl (pH 8.3), 50 mM potassium chloride, 200 µg/mL of bovine serum albumin (BSA), 1.5 mM magnesium chloride, 200 µM of each dNTP, 200 nM of each primer and 0.025 U/µL of thermostable *Taq* polymerase. A touch-down amplification protocol was run on a Perkin Elmer GeneAmp PCR System 9600 thermocycler. The first two cycles included a denaturing step at 94 °C for 30 s, an annealing step at 60 °C for 30 s and an extension step at 72 °C for 30 s. The next 11 cycles consisted of a denaturing step at 94 °C for 15 s, an annealing step at 60 °C for 15 s (which was subsequently decreased by 0.5 °C every cycle until a final annealing temperature of 54 °C was reached) and an extension step at 72 °C for 15 s. Conditions for the last 25 cycles were: denaturation at 94 °C for 15 s, annealing at 54 °C for 15 s and extension at 72 °C for 15 s, followed by a final extension at 72 °C for 3 min. Five microlitres of the amplification products was examined on a horizontal 1.5% agarose gel in 0.5× TBE (Tris-Borate-EDTA), containing 0.2 µg/mL of ethidium bromide, and photographed using a Polaroid MP-4 land camera with UV illumination.

Sequencing gels and resolution of microsatellites

Based on the intensity of the product band, the rest of the product (5 µL) was diluted with 10–30 µL of stop solution (95% deionized formamide, 10 mM sodium hydroxide, 0.05% Bromophenol blue, 0.05% xylene cyanol). Samples were denatured at 95 °C for 3 min before loading 3–3.5 µL onto pre-electrophoresed vertical denaturing polyacrylamide gels. The composition of the 0.4 mm × 35 cm × 45 cm gels was: 6% acrylamide (19 : 1 acrylamide:bisacrylamide), 8 M urea and 1× TBE. The gel mix was deionized with 150 mg/100 mL of Amberlite Mixed Bed ion-exchange resin (Sigma MB-1) for 30–60 min, following which it was filtered through 0.22-µm syringe filters to remove particulate matter. For glass plate preparation, Acrylase (Stratagene Cloning Systems) was used for a nonstick coating on one plate and gamma-methacryloxypropyltrimethoxysilane (Promega) for gel binding treatment on the other. The upper reservoir buffer was 1× TBE and the lower reservoir buffer was 0.5 × TBE and 0.5 M sodium acetate (pH 7.0). The gels were electrophoresed for 2.5–3.5 h at a constant temperature

of 50 °C with 110 W of limiting power. After electrophoresis, the gels were stained with silver, according to the Promega Silver Sequence protocol, with some modifications. All solutions were freshly prepared using double glass-distilled water (ddH₂O) and all steps were performed at room temperature, but the developer solution was prechilled to 4–6 °C to reduce background staining. After electrophoresis, the gel bound to a glass plate was fixed for 20–35 min (until all the tracking dye was no longer visible) in 10% acetic acid and then rinsed three times, 2 min each, with ddH₂O. The used fixing solution was stored at 4 °C for later use. The gels were then stained for 30 min with 6 mM silver nitrate containing 0.056% formaldehyde and then rinsed with ddH₂O for 8–10 s. For development of the stain, 800 mL of chilled developer (0.28 M sodium carbonate, 0.056% formaldehyde, 8 µM sodium thiosulphate) was added with gentle agitation and decanted off when the first bands became visible. An additional 1200 mL of developing solution was added and agitation continued until the development was nearly complete. The development reaction was stopped by adding 1200 mL of chilled, previously used, fixing solution and the gel was agitated for another 3 min. The gels were finally washed twice (5 min each) with ddH₂O and air-dried overnight. Permanent records of the images were made by producing contact prints on APC film (Promega Corp.). Super low ladders of 100 bp and 20 bp (GenSura Lab., Inc.) were used as size standards.

The gels were scored for microsatellite DNA genotypes of individual trees. The alleles at a SSR locus were identified by their molecular size and were designated alphabetically from larger to smaller molecular size.

Data analysis

Standard genetic diversity parameters (allele frequencies, allelic complements, per cent polymorphic loci, average number of alleles per locus) and Nei's (1978) unbiased estimates of mean expected and observed heterozygosity for pre- and postharvest stands from the SSR data were calculated using BIOSYS-1 (Swofford & Selander 1989). Genotype and allelic data were used also to calculate nontraditional genetic diversity measures. These include G_{M_r} genotype multiplicity, the multiplication product of all genotype combinations across all loci (Bergmann *et al.* 1990); V , genic diversity, which is the harmonic mean of the effective number of alleles per locus taken over all loci (Bergmann *et al.* 1990); V_{gamv} or hypothetical gametic diversity, which is the multiplication product of all the number of gametes theoretically available under linkage equilibrium (Gregorius 1987; Bergmann *et al.* 1990); and LP, latent genetic potential, which is the difference between total number of alleles and effective number of alleles summed over all loci (Bergmann *et al.* 1990). In

addition, expected and observed genotype additivity, G_A (Rajora 1996), was calculated by summing the theoretical number of single-locus genotypes over all loci which could be expected from the allelic complement under Hardy–Weinberg expectations and by summing the actual single-locus genotypes present, respectively.

Alleles were assigned to one of the four frequency classes: high ($P \geq 0.75$); intermediate ($0.75 > P \geq 0.25$); low ($0.25 > P \geq 0.01$); and rare ($P < 0.01$). For comparison, alleles were also classified as common ($P \geq 0.05$) and rare ($P < 0.05$), as suggested by Marshall & Brown (1975).

With a bottleneck in population size, genetic diversity is expected to decrease. The number of alleles expected to be retained after the harvesting bottleneck in each of the two stands was calculated as described in Frankel & Soule (1981). A paired *t*-test and a χ^2 -test were performed to test the differences between the observed and expected number of alleles in the postharvest gene pool. The expected postharvest heterozygosity was calculated according to the drift theory. Drift depletes genetic variance (heterozygosity) at the rate of $1/2N_e$ per generation, where N_e is the effective population size.

Results

Twelve microsatellite DNA loci were polymorphic, and one locus (*RPS61*) was found to be monomorphic in all 222 individuals examined (Table 1). Two to 21 alleles were detected at a polymorphic locus, with a total of 132 alleles at the 12 polymorphic loci in both stands. Microsatellite DNA profiles (at locus *RPS50*) of some white pine trees from stands A and B are shown in Fig. 1. Both Galloway Lake stands had high microsatellite genetic diversity (Table 2), which was $\approx$ fourfold that of allozyme genetic diversity (Buchert *et al.* 1997). Stand A was genetically more diverse than stand B for allelic richness, whereas stand B was more diverse than stand A for heterozygosity. Stand A had a greater number of rare alleles than stand B (Table 3) and rare alleles do not contribute significantly to heterozygosity estimates. The distribution of microsatellite alleles in four different frequency classes was different from that of allozyme alleles, with 88 and 90% of alleles in low and rare frequency classes, respectively (Table 3). When classified into two classes (Marshall & Brown 1975), the alleles were almost equally distributed between the 'common' and 'rare' classes (Table 3). There were 12 alleles unique to stand A and seven unique to stand B. One allele at the *RPS50* locus, unique to stand B, is shown in Fig. 1(B). Removal of $\approx 75\%$ of white pine trees substantially changed the frequencies of a number of alleles in the residual gene pool (data not shown).

The genetic diversity was substantially reduced in the postharvest residual gene pools and the reductions were

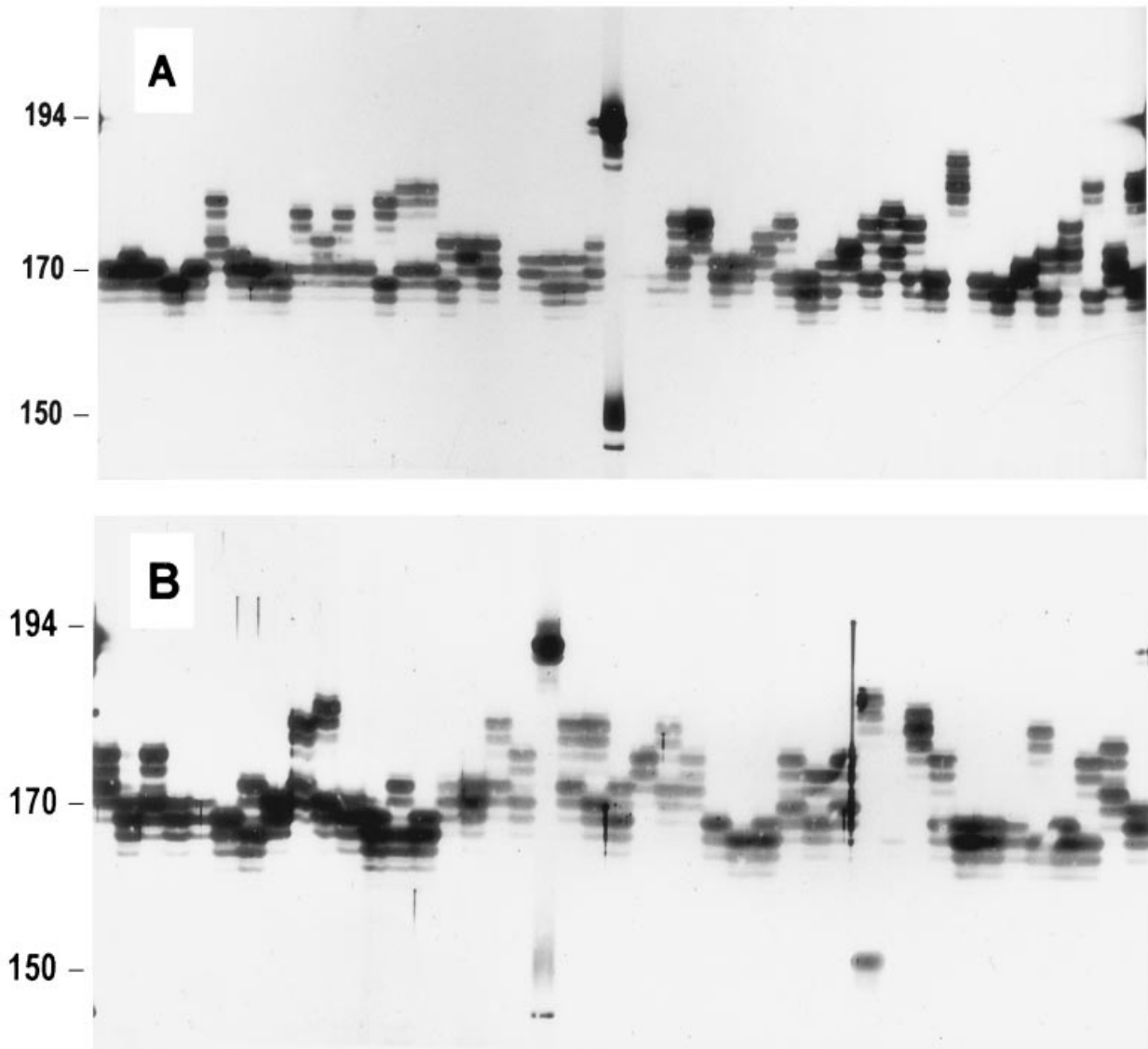


Fig. 1 Microsatellite DNA profiles of 48 white pine trees from stand A (A) and 42 trees from stand B (B) for the locus *RPS50*. One individual in stand B has a rare and stand-specific allele.

Table 2 Measures of microsatellite DNA genetic diversity in preharvest intact and postharvest residual gene pools, Galloway Lake old-growth white pine*

Diversity parameter	Stand A		Stand B	
	Preharvest	Postharvest	Preharvest	Postharvest
No. of trees	120	30	102	22
Total no. of alleles over 13 loci	125	93	120	89
Mean no. of alleles per locus	9.62 (1.83)	7.15 (1.29)	9.23 (1.76)	6.85 (1.18)
Mean effective no. of alleles per locus	4.24 (0.89)	4.05 (0.85)	4.80 (1.05)	3.82 (0.72)
No. of polymorphic loci	12	12	12	12
Per cent loci polymorphic	92.3	92.3	92.3	92.3
Mean heterozygosity (observed)	0.505 (0.066)	0.485 (0.072)	0.538 (0.068)	0.535 (0.073)
Mean heterozygosity (expected)	0.599 (0.079)	0.587 (0.081)	0.615 (0.081)	0.585 (0.079)

*SE in parentheses.

Table 3 Distribution of alleles in allele frequency classes in preharvest and postharvest (residual) old-growth white pine stands

Allele	Number of alleles (percentage of total number of alleles)					
frequency	Stand A			Stand B		
class	Postharvest (residual)			Postharvest (residual)		
	Preharvest	Actual*	Retained† from preharvest stand	Preharvest	Actual*	Retained† from preharvest stand
Total no. of alleles	125	93	93	120	89	89
Four classes						
High $P \geq 0.75$	2 (1.6)	4 (4.3)	2 (2.2)	3 (2.5)	4 (4.5)	3 (3.4)
Intermediate						
$0.75 > P \geq 0.25$	13 (10.4)	10 (10.8)	13 (14.0)	9 (7.5)	10 (11.2)	9 (10.1)
Low						
$0.25 > P \geq 0.01$	89 (71.2)	79 (84.9)	73 (78.5)	96 (80.0)	75 (84.3)	76 (85.4)
Rare $P < 0.01$	21 (16.8)	0 (0)	5 (5.4)	12 (10.0)	0 (0)	1 (1.1)
Two classes						
Common						
$P \geq 0.05$	53 (42.4)	38 (40.9)	52 (55.9)	60 (50.0)	43 (48.3)	56 (62.9)
Rare $P < 0.05$	72 (57.6)	55 (59.1)	41 (44.1)	60 (50.0)	46 (51.7)	33 (37.1)

*Alleles were present in these frequency classes in the postharvest gene pool.

†Alleles were present in these frequency classes in the preharvest gene pool and were retained in the postharvest gene pool, but not necessarily in the same frequency classes.

Table 4 Allele losses in the residual old-growth white pine stands after harvesting

Allele losses	Stand A	Stand B
Total number (%)	32 (25.6)	31 (25.8)
Four allele frequency classes*		
Number (%) rare	16 (76.2)	11 (91.7)
Number (%) low	16 (18.0)	20 (20.8)
Number (%) intermediate and high	0 (0)	0 (0)
Alleles lost of total alleles lost in different frequency classes (%)		
Rare	50.0	35.5
Low	50.0	64.5
Intermediate and high	0.0	0.0
Alleles lost of total alleles detected (%)		
Rare	12.8	9.2
Low	12.8	16.7
Two allele frequency classes†		
Number and (%) rare	31 (43.1)	27 (45.0)
Rare alleles lost of the total alleles (%)	24.8	22.5
Number and (%) common	1 (1.9)	4 (6.7)
Common alleles lost of the total alleles (%)	0.8	3.3

*High $P \geq 0.75$; intermediate $0.75 > P \geq 0.25$; low $0.25 > P \geq 0.01$; rare $P < 0.01$.

†Common $P \geq 0.05$; rare $P < 0.05$.

fairly uniform between the stands (Tables 2–6). With a reduction in the breeding population of $\approx 75\%$, $\approx 26\%$ of the alleles were lost from the residual gene pools of both stands (25.6 and 25.8% in stands A and B, respectively) after harvesting. However, heterozygosity was reduced by only $\approx 1\text{--}5\%$ in the postharvest residual gene pool

(Table 2). All the allele reductions occurred in the low and rare frequency classes (Table 4) when alleles were classified into four classes. With alleles classified into only two classes, in addition to loss of rare alleles, one and four alleles were lost from the common frequency class from stands A and B, respectively (Table 4). Only five out of 12

Table 5 Number of alleles unique and common to old-growth white pine stands A and B and retained in residual trees after harvesting

Alleles	Number of alleles and (%)		
	Total	Retained after harvesting in the residual trees	Lost after harvesting in the residual trees
Unique to stand A*	12 (9.1)	5 (3.8)	7 (5.3)
Unique to stand B*	7 (5.3)	4 (3.0)	3 (2.3)
Common to stands A and B*	113 (85.6)		
Retained in residual stands A and B*		70 (53.0)	
Lost from residual stands A and B*			10 (7.6)
Lost from residual stand A only†			15 (12.0)
Lost from residual stand B only‡			18 (15.0)

*Percentage of 132 alleles detected in both stands A and B.

†Percentage of 125 alleles detected in stand A.

‡Percentage of 120 alleles detected in stand B.

Parameter*	Stand A		Stand B	
	Preharvest	Postharvest	Preharvest	Postharvest
LP	69.91	40.29	57.66	39.30
G_M (expected)	1.153×10^{19}	3.658×10^{16}	6.629×10^{18}	1.917×10^{16}
G_M (observed)	8.936×10^{10}	4.313×10^9	6.909×10^{10}	3.082×10^9
V	2.50	2.42	2.60	2.41
V_{gam}	4 214 001	2 626 847	12 314 320	1 884 498
G_A (expected)	924	509	855	458
G_A (observed)	276	135	265	100

* G_A , genotype additivity (Rajora 1996); G_M , multilocus genotype multiplicity (Bergmann *et al.* 1990); LP, latent genetic potential (Bergmann *et al.* 1990); V , genic diversity (Gregorius 1987); V_{gam} , hypothetical gametic diversity (Gregorius 1987).

Table 6 Latent genetic potential and non-traditional genetic diversity measures for the preharvest intact and postharvest residual old-growth white pine stands

alleles unique to stand A were retained in the post-harvest stand A, whereas four out of seven unique alleles were retained in postharvest stand B (Table 5). Out of the 113 alleles common to both stands, 10 were lost from both residual stands, 15 from residual stand A and 18 from residual stand B (Table 5).

The nontraditional genetic diversity measures were also substantially reduced in the residual gene pool as a result of harvesting (Table 6) and, as with most of the standard genetic diversity parameters (Table 2), both stands responded similarly. Owing to allelic losses, expected and observed postharvest G_M values represented a very small percentage of preharvest values. The reductions in the other nontraditional genetic diversity parameters were as follows: latent genetic potential –42 and 32%; genotype additivity (observed) –51 and 62%; genotype additivity (expected) –45 and 46%; gametic diversity –38 and 85% (in stands A and B, respectively) (Table 6).

The expected number of total alleles in the postharvest

residual gene pool was 97 for stand A and 91 for stand B. This number was significantly different (higher) from the observed number of alleles at the 5% significance level based on a paired *t*-test for both stands. However, the differences were not significant based on the χ^2 -test. Heterozygosity values expected under the drift theory in the postharvest residual stands were as follows: 0.497 and 0.526 for observed heterozygosity, and 0.589 and 0.601 for expected heterozygosity in stands A and B, respectively. Thus, the actual heterozygosity values in the postharvest gene pools (Table 2) were very similar (+1.7% to –2.7% difference) to these postbottleneck expected values.

Discussion

Our study demonstrates a reduction in allelic richness of $\approx 26\%$ in the postharvest residual gene pools of two old-growth white pine stands. All the allelic losses were in the low-frequency and rare alleles in both stands, when alleles were classified into four classes. Low-frequency

allele losses are more significant because this allele class accounted for 71 (stand A) and 80% (stand B) of all alleles assayed, which was the highest proportion of the allelic complement in the preharvest stands. In the present study, low and rare frequency alleles accounted for $\approx 90\%$ of the total allelic complement. Although the genetic makeup of each of the stands examined was different, response to the harvesting bottleneck was relatively uniform in both stands. This suggests that genetic diversity reductions, as observed in the present study, may be common in the residual stands when forests of this type are harvested at these intensities.

Our study was based on a total census of the white pine trees in the pristine preharvest and residual postharvest gene pools. Thus, the results presented here represent the true picture of events instead of the conclusions drawn from random sampling; therefore, no sampling error is likely to be associated with the results.

Although the number of white pine individuals analysed was higher for microsatellite DNA analysis (122 in stand A and 102 in stand B) than analysed for allozymes in an earlier report (96 in stand A and 95 in stand B; Buchert *et al.* 1997), the results for genetic implications of harvesting were very similar, and in some cases almost identical, for both analyses. Allozyme analysis of the same stands indicated that the allelic richness was reduced by $\approx 25\%$, and the reductions in other genetic diversity parameters were in the range of 25–50% (Buchert *et al.* 1997). Thus, our microsatellite DNA results support and reconfirm our earlier findings from allozyme analysis.

Results from the present study also confirm our observations from allozyme studies (Buchert *et al.* 1997) that for pristine old-growth white pine populations, allelic richness measures are more useful than allelic evenness measures (e.g. heterozygosity) when quantifying effects of perturbations on gene pools (Marshall & Brown 1975; Leberg 1992). Although heterozygosity measures are widely used descriptors of genetic diversity, in our study they did not follow the reduction in allelic complements caused by harvesting. Heterozygosity is a measure of allelic evenness, which is not as sensitive to changes resulting from harvesting bottlenecks as are allelic-richness measures. Drift depletes genetic variance (heterozygosity) very slowly at the rate of $1/2N_e$ per generation, where N_e is the effective population size. Theoretically, two individuals could maintain 50% of the original heterozygosity of a large population. Therefore, populations with a moderate number of individuals could lose little of expected heterozygosity in one generation. In fact, heterozygosity has even been found to increase in some postbottleneck populations (Leberg 1992).

Genetic diversity is reduced with bottleneck in the population size (Frankel & Soule 1981). The reductions in microsatellite DNA genetic diversity in the postharvest

gene pool were slightly higher than, but more or less followed, theoretical expectations. Thus, our study provides empirical data to support the population genetic theory of drift for genetic diversity reductions after population bottleneck.

The number of private or unique alleles in each stand for microsatellite DNA markers is lower than that observed for the allozyme markers but relatively high when compared to levels reported for other white pine populations (Beaulieu & Simon 1994; Rajora *et al.* 1998) and other species (Neale 1985; Yazdani *et al.* 1985; Cheliak *et al.* 1988; El-Kassaby & Ritland 1996). We think that this may be a result of the total census of the stands rather than random sampling of a smaller number of individuals. The loss of private alleles from both stands may be of concern for the integrity of locally adapted gene pools after harvesting. Microsatellite markers have generally been considered to be selectively neutral. However, recent information suggests that the microsatellite sequences primarily reside in the single-copy region of the pine genome (Williams 1999). Thus, certain microsatellite makers may have selective value or be linked to genes of selective value in the genome.

If the microsatellite loci are a part of, or have linkage to, the genes that play a part in fitness and adaptability, the reductions in allelic richness (as observed in our study) have great significance. Under an adaptive gene action hypothesis, the low and rare frequency allozyme alleles in a population represent much of the genetic potential required for population adaptation to long-term environmental changes, because alleles of higher frequencies have probably been selected for current or recent-past environments. Low levels of latent genetic potential have been correlated with forest decline in northern and central populations of European silver fir (*Abies alba* Mill.), while high levels appear to be correlated with healthy southern European populations (Bergmann *et al.* 1990). The latent genetic potential of the postharvest residual gene pools was reduced to $\approx 60\%$ of that initially present in the preharvest gene pools. This reduction is approximately equivalent to that reported in northern and central populations, in contrast to southern populations of European silver fir (Bergmann *et al.* 1990). Reductions in the levels of $G_{M'}$, V_{gam} and G_A may also result in reduced long-term evolutionary potential of the populations. These reductions are comparable to the differences found between Norway spruce (*Picea abies* [L.] Karst.) populations that were tolerant and sensitive to airborne pollutants (Raddi *et al.* 1994), and to differences between pollution-tolerant and -sensitive beech clones (*Fagus sylvatica* L.) (Müller-Starck 1985).

The selective value of the affected alleles is not known. As the sample stands are not yet well regenerated, we have not been able to determine to what extent the

genetic diversity of the residual parental stand is passed on to the regenerated stand. If some of the rare alleles in white pine are deleterious, there may be some eugenic effects from eliminating 35 and 50% of the rare alleles from the breeding population. However, it is also possible that these rare alleles represent genetic repertoire that may be useful for future adaptations to altered disease and environmental conditions. There is evidence for the adaptive value of certain rare alleles in environmentally stressed beech populations (Müller-Starck 1985). In light of the need to maintain genetic diversity for long-term evolution in changing environments, and recognizing our current lack of understanding of the adaptive value of these low and rare frequency alleles, it seems prudent to ensure that the integrity of gene pools be maintained, and that natural selection be allowed to remove deleterious alleles during the life cycle of the progeny generation.

The real genetic losses in the white pine stands studied will be if the allelic losses observed in the postharvest residual gene pool are passed on to the next generation. In a partial cutting system, as used in our study, the residual seed trees provide the seed source for postharvest regeneration of the stand. The seed source could also be supplemented by seed dispersal from outside the stand and the seed from the ground seed bank, if any, in the regenerating stands. However, there was very little or no regeneration even 3 years after the harvesting of these stands, which indicates little or no contribution from seed dispersal or seed bank. We do not know to what extent gene migration will ameliorate the genetic losses measured in these postharvest residual stands. Estimated gene flow was relatively high (≈ 10 migrants per generation) between the studied stands and is also reported to be very high in other eastern white pine populations (Beaulieu & Simon 1994; Rajora *et al.* 1998). Therefore, it is possible that genetic diversity in naturally regenerated populations may be as high as preharvest parental stands owing to pollen migration from surrounding stands. However, our observations from the spatial genetic structure in these stands (data to be presented elsewhere) suggest that white pine gene dispersal is rather limited in space. Also, all surrounding stands within 3 km of the study stands have been harvested in a similar manner, so reintroduction of lost alleles from surrounding stands may be problematic.

The effective breeding population of each residual stand will probably be actually smaller than the census population because some of the residual trees are effectively noncontributors to the gene pool. We do not know how the changes in stand density will affect mating systems of the residual trees and, in turn, how the remaining genetic diversity will be passed on to progeny populations. It is possible that even greater genetic diversity losses may result in stands where current harvesting policies reduce residual stand numbers below the approximate 50

stems per hectare as experienced in these study stands. However, eastern white pine seems to be resilient to genetic-diversity losses as a result of small population size and forest fragmentation (Rajora *et al.* 1998).

Both our previous allozyme and the present microsatellite DNA studies suggest that in order to maintain the genetic integrity of these remaining intact old-growth gene pools, these stands must be carefully regenerated before any significant genetic erosion occurs as a result of anthropogenic activity.

Acknowledgements

The study was funded by the Natural Sciences and Engineering Research Council of Canada through a Strategic Research Grant (STR 0167646) to Bruce P. Dancik, Om P. Rajora and George P. Buchert. We thank V. Ball and Doug Offord for help in locating the study sites and assistance with field work; Craig Echt for providing some eastern white pine microsatellite DNA markers; Derek Chao for DNA extraction; Renato Vitic for technical assistance with microsatellite DNA analysis; and Dr Alex Mosseler for his review of the manuscript. The manuscript also benefited from the comments and suggestions of Dr Loren Rieseberg and two anonymous reviewers.

References

- Barnes BV (1989) Old-growth forests of the northern lake states, a landscape ecosystem perspective. *Natural Areas Journal*, **9**, 45–57.
- Beaulieu J, Simon J-P (1994) Genetic structure and variability in *Pinus strobus* in Quebec. *Canadian Journal of Forest Research*, **24**, 1726–1733.
- Bergmann F, Scholz F (1987) The impact of air pollution on the genetic structure of Norway spruce. *Silvae Genetica*, **36**, 80–83.
- Bergmann F, Gregorius H-R, Larsen JB (1990) Levels of genetic variation in European silver fir (*Abies alba*) – are they related to the species' decline? *Genetica*, **82**, 1–10.
- Buchert GP (1994) Genetics of white pine and implications for management and conservation. *Forestry Chronicle*, **70**, 427–434.
- Buchert GP, Rajora OP, Hood JV, Dancik BP (1997) Effects of harvesting on genetic diversity in old-growth eastern white pine in Ontario, Canada. *Conservation Biology*, **11**, 747–758.
- Carleton TJ, Gordon AM (1992) Understanding old-growth red and white pine dominated forests in Ontario. *Forest Landscape Ecology Series 2*, Ontario Forest Research Institute, Sault Ste. Marie, Ontario.
- Cheliak WM, Murray G, Pitel JA (1988) Genetic effects of phenotypic selection in white spruce. *Forest Ecology and Management*, **24**, 139–149.
- Echt CS, Nelson CD (1997) Linkage mapping and genome length in eastern white pine (*Pinus strobus* L.). *Theoretical and Applied Genetics*, **94**, 1031–1037.
- Echt CS, May-Marquardt P, Hsieh M, Zahorchak R (1996) Characterization of microsatellite markers in eastern white pine. *Genome*, **39**, 1102–1108.
- El-Kassaby YA, Ritland K (1996) Impact of selection and breeding on the genetic diversity in Douglas-fir. *Biodiversity and Conservation*, **5**, 795–813.

- Frankel OH, Soule ME (1981) *Conservation and Evolution*. Cambridge University Press, Cambridge.
- Gregorius H-R (1987) The relationship between the concepts of genetic diversity and differentiation. *Theoretical and Applied Genetics*, **74**, 397–401.
- Gregorius H-R, Hattermer HH, Bergmann F, Müller-Starck G (1985) Umweltbelastung und Anpassungsfähigkeit von Baumpopulationen. *Silvae Genetica*, **34**, 230–241.
- Knowles P (1985) Comparison of isozyme variation among natural stands and plantations: jack pine and black spruce. *Canadian Journal of Forest Research*, **15**, 902–908.
- Leberg PL (1992) Effects of population bottlenecks on genetic diversity as measured by allozyme electrophoresis. *Evolution*, **46**, 477–494.
- Ledig FT (1988) The conservation of diversity in forest trees. *Bioscience*, **38**, 471–479.
- Ledig FT (1992) Human impacts on genetic diversity in forest trees. *Oikos*, **63**, 87–108.
- Marshall DR, Brown AHD (1975) Optimum sampling strategies in genetic conservation. In: *Crop Genetic Resources for Today and Tomorrow* (eds Frankel OH, Hawkes JG), pp. 53–80. Cambridge University Press, London.
- Mosseler A, Rajora OP (1998) Monitoring population viability in declining tree species using indicators of genetic diversity and reproductive success. In: *Environmental Forest Science* (ed. Sassa K), pp. 333–344. Kluwer Academic Publishers, Dordrecht, the Netherlands.
- Müller-Starck G (1985) Genetic differences between 'tolerant' and 'sensitive' beeches (*Fagus sylvatica* L.) in an environmentally stressed adult forest stand. *Silvae Genetica*, **34**, 241–247.
- Müller-Starck G, Baradat Ph, Bergmann F (1992) Genetic variation within European tree species. *New Forests*, **6**, 23–47.
- Namkoong G (1991) Biodiversity — issues in genetics, forestry and ethics. *Forestry Chronicle*, **68**, 438–443.
- Neale DB (1985) Genetic implications of shelterwood regeneration of Douglas-fir in southwestern Oregon. *Forest Science*, **31**, 995–1005.
- Nei M (1978) Estimation of average heterozygosity and genetic distance from a small number of individuals. *Genetics*, **89**, 583–590.
- Oleksyn J, Prus-Glowacki W, Giertych M, Reich PB (1994) Relation between genetic diversity and pollution impact in a 1912 experiment with east European *Pinus sylvestris* provenances. *Canadian Journal of Forest Research*, **24**, 2390–2394.
- Perera AH, Baldwin DJ (1993) Spatial characteristics of eastern white pine and red pine forests of Ontario. *Forest Landscape Ecology Series* 9. Ontario Forest Research Institute, Sault Ste. Marie, Ontario.
- Raddi S, Stefanini FM, Camussi A, Giannini R (1994) Forest decline index and genetic variability in *Picea abies* (L.) Karst. *Forest Genetics*, **1**, 33–40.
- Rajora OP (1996) Effects of forestry practices on genetic diversity: implications for sustainable forest management and gene conservation. *Plenary presentation, IUFRO conference, Diversity and adaptation in forest ecosystems in a changing world, August 5–10, 1996*. University of British Columbia, Vancouver, B. C.
- Rajora OP (1999) Genetic biodiversity impacts of silvicultural practices and phenotypic selection in white spruce. *Theoretical and Applied Genetics*, **99**, 954–961.
- Rajora OP, DeVerno L, Mosseler A, Innes D (1998) Genetic diversity and population structure of disjunct Newfoundland and central Ontario populations of eastern white pine (*Pinus strobus* L.). *Canadian Journal of Botany*, **76**, 500–508.
- Swofford DL, Selander RB (1989) *BIOSYS-1. A computer program for the analysis of allelic variation in population genetics and biochemical systematics. Release 1.7*. Illinois Natural History Survey, Illinois.
- Welch D, Clark T, Clark K (1992) Fauna of red and white pine old-growth forests in Ontario, issues and recommendations for research. *Forest Landscape Ecology Series* 3. Ontario Forest Research Institute, Sault Ste. Marie, Ontario.
- Williams C (1999) The peculiarities of pine genome. *Plant and Animal Genome VII Conference*. January 17–21, 1999, San Diego, California.
- Yazdani R, Muona O, Rudin D, Szmidt AE (1985) Genetic structure of a *Pinus sylvestris* L seed-tree stand and naturally regenerated understory. *Forest Science*, **31**, 430–436.

The work presented here is a part of a University of Alberta and Ontario Ministry of Natural Resources collaborative strategic research project. Om Rajora is an Associate Research Professor of Forest Genetics and Biotechnology, and his research focus is forest molecular, population, conservation and evolutionary genetics, genomics and genetic effects of forest management practices and forest fires. Muhammad Rahman is a postdoctoral research fellow. George Buchert is a Physiological Geneticist, with research interest in population genetics and genetic implications of forest management practices. Bruce Dancik is a Professor of Forest Genetics and Associate Vice-President (Academic), with research interests in molecular, population and evolutionary genetics.
